

Product Carbon Footprint Study Report

Product Name: transistor production, surface-mounted

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: transistor production, surface-mounted

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 kg transistor production, surface-mounted

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1kg transistor production, surface-mounted.

2. System Boundary

transistor production, surface-mounted, in China, in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of transistor production, surface-mounted produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **27.039 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

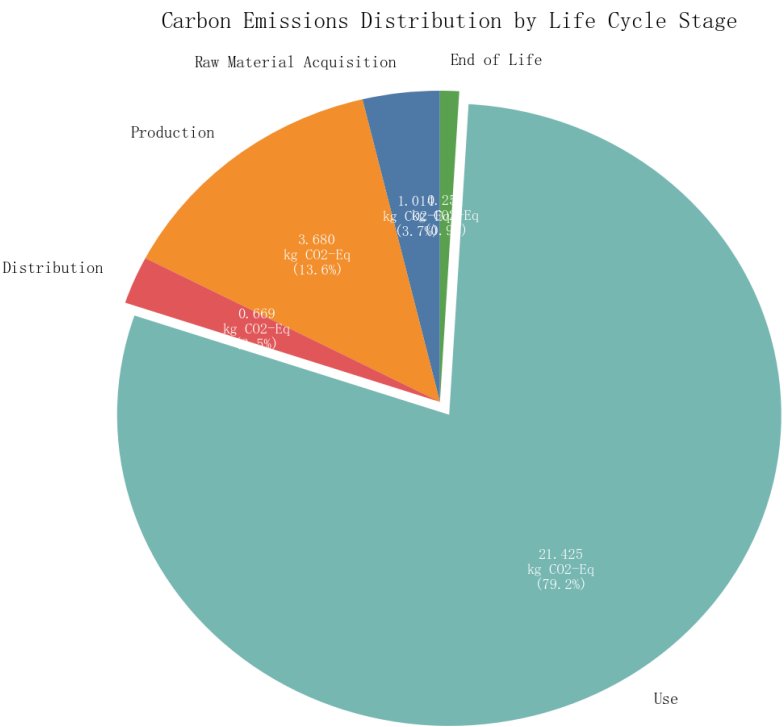


Table 1: Carbon Emissions by Life Cycle Stage

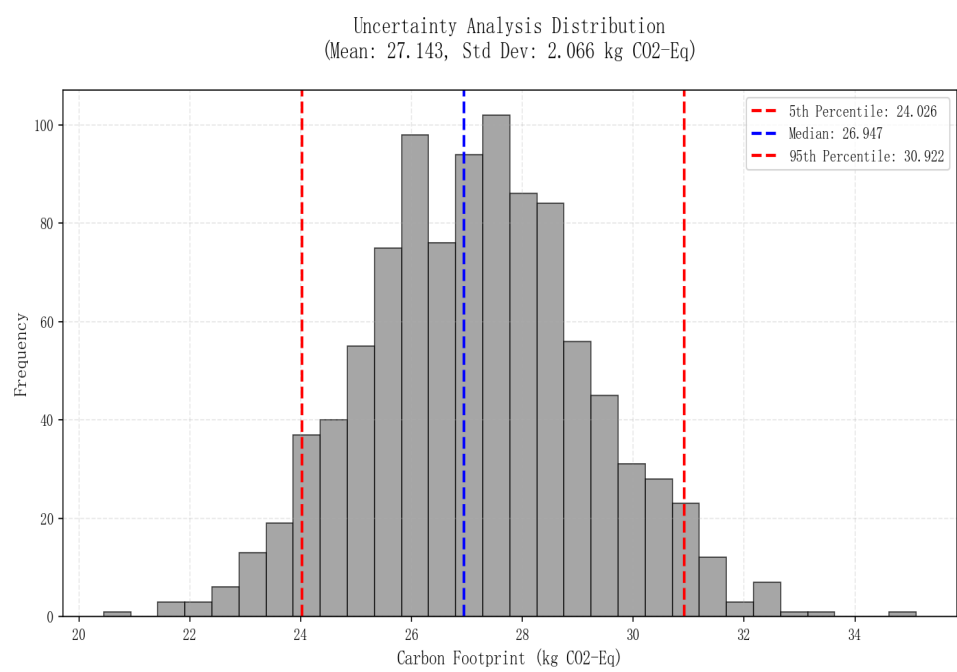
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	1.014	3.75
Production	3.680	13.61
Distribution	0.669	2.47

Use	21.425	79.24
End of Life	0.251	0.93
Total	27.039	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO2-Eq)
Mean	27.143
Standard Deviation	2.066
Coefficient of Variation (CV)	7.61%
Median	26.947
5th Percentile	24.026
95th Percentile	30.922

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the surface-mounted transistor is estimated at 27.039 kg CO₂-Eq per functional unit. This represents a moderate-to-high impact profile for a discrete semiconductor component, largely shaped by the device's operational behavior over its lifetime. The lifecycle stage assessment reveals a strongly use-phase-dominated pattern: the use stage contributes 21.425 kg CO₂-Eq (79.2%), followed by production at 3.680 kg CO₂-Eq (13.6%) and raw material acquisition at 1.014 kg CO₂-Eq (3.7%). Distribution and end-of-life play only minor roles, accounting for 2.5% and 0.9% of the total, respectively.

The overwhelming dominance of the use phase indicates that electricity consumption during the transistor's service life is the primary environmental burden, exceeding manufacturing-related impacts by roughly a factor of six. This outcome is typical for electronic components operating under continuous or prolonged load conditions. In contrast, the combined upstream chain (raw material plus production) represents just 17.3%, suggesting that material efficiency and clean manufacturing—while relevant—are secondary to in-field energy performance. The negligible end-of-life share reflects limited recycling or disposal burdens under the current model assumptions.

2. Statement of Assumptions and Limitations

This assessment was conducted under a cradle-to-grave system boundary covering raw material extraction, manufacturing, distribution, use, and end-of-life. A central assumption concerns the use-phase scenario, including the transistor's operational power draw, annual operating hours, and service lifetime typical of consumer electronics. Monte Carlo simulation with 1,000 iterations yielded a mean of 27.143 kg CO₂-Eq with a coefficient of variation of 7.61%, and a 90% confidence interval of 24.026–30.922 kg CO₂-Eq, indicating that the result is statistically stable within roughly $\pm 14\%$ of the mean.

Several limitations must be acknowledged. The top process contribution analysis returned "Unknown" entries with zero values for all five leading processes, meaning that process-level disaggregation was either unavailable or masked through aggregation. This considerably constrains hotspot identification and prevents verification of the underlying unit-process data. Secondary data sources, likely drawn from generic life cycle inventory databases, may not fully capture supplier-specific manufacturing conditions, regional energy mixes, or actual transport routes. Furthermore, system boundaries may exclude capital equipment, packaging, and ancillary operational emissions, potentially leading to a slight underestimation of the total footprint.

3. Suggestions for Improvement

- ****Prioritize use-phase efficiency improvements.**** Since the use stage drives 79.2% of the total footprint, focus engineering efforts on reducing on-state resistance, switching losses, and standby power consumption. Publish clear energy-efficiency specifications and guide customers toward operating profiles that minimize electricity demand over the product's lifetime.

- ****Decarbonize the production stage.**** Transition wafer fabrication and assembly facilities to renewable electricity sources (e.g., solar, wind, or hydropower) and improve the energy efficiency of cleanrooms, furnaces, and ion implantation equipment. A 20% reduction in production-stage emissions would save approximately 0.74 kg CO₂-Eq per unit.
- ****Optimize raw material selection and supply chains.**** Substitute high-impact materials, such as replacing gold bonding wire with copper or palladium alternatives, and source substrates and chemicals from regional suppliers. Localizing the supply chain would also reduce the distribution share (2.5%) and associated transport emissions.
- ****Strengthen end-of-life circularity.**** Implement take-back programs and design the transistor package for easier solder separation and material recovery. Although end-of-life currently contributes only 0.9%, improving recyclability can convert this stage into a net-positive loop that recovers metals and reduces primary material demand.
- ****Close data quality gaps.**** Replace the "Unknown" process entries with disaggregated, supplier-specific unit processes. Document the electricity mix, transport distances, and waste treatment routes explicitly, and conduct a sensitivity analysis on use-phase assumptions to further reduce the coefficient of variation (currently 7.61%) and increase confidence in decision-making.